EE1007
Introduction to Computer Science Laboratory

PA4
Myspice

Andy, Yu-Guang Chen
Assistant Professor, Department of EE
National Central University
andyygchen@ee.ncu.edu.tw

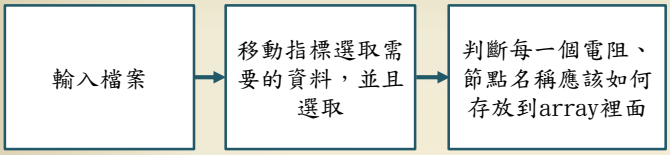
Main function

◆You can devide this problem into 5 functions.

```
*** Welcome to MySpice ***
============================================
Please type 1 to input file
Please type 2 to calculate voltage and current
Please type 3 to swap stage
Please type 4 to merge stage
Please type 5 to output file
Please type 6 to exit
Enter your selection Here: 1
```

1. Input file

◆思考方向:




```
1  stage 2
2
3  v1 vdd gnd 5V
4
5  RS1_1    vdd ns1_1    2K
6  RS1_2    ns1_1   n1   2K
7  RP1_1    n1   gnd 2K
8  RS2_1    n1   n2   2K
9  RP2_1    n2   np2_1    3K
10 RP2_2    np2_1    gnd 3K
11
12 end
```


1. Input file

◆How to get input information:

```
void input_file(int stages, int voltage, int resistor[15][], string node_name[][]){
    char data[];

//input file
    ifstream infile();
    seekg();
    infile >> data;
    stages = atoi(data[]);

//voltage source
    seekg();
    infile >> data;
    voltage = atoi(data[]);

//initialize resistor & node name
    seekg();
    while(){
        infile >> data;
        resistor[][] = ;
        node_name[][] = ;
        resistor_num = ;

    }
}
```

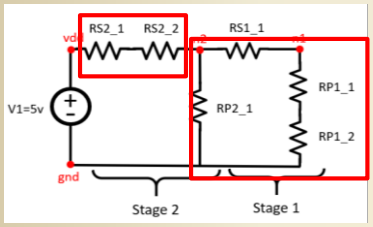
2. calculate voltage and current

◆思考方向:

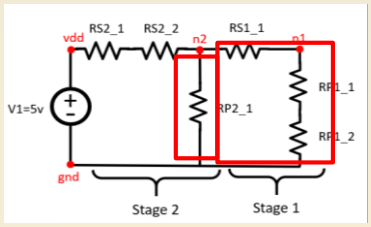
先得出最簡化的電路，只有一個電壓源跟一個等效電阻，就可以求出總電流了 → 算出對每個node往後看得等效電阻可以簡化問題 → 利用分流分壓公式、算出壓降等等方式 → 得出電流電壓

2. calculate voltage and current

◆思考方向:



看成串聯兩個等效電阻
可以算分壓



看成並聯兩個等效電阻
可以算分流

2. calculate voltage and current

◆How to get calculate voltage and current :

```
void calculate(int stages, int voltage, int resistors[], int resistor_num[], int V_node[], int I_resistor[][]){
    int total_i;
    int eq_resistor[15][2];

//the equivalent resistor
    for(int i=0; i< stages; i++){
        eq_resistor[][] = ;
    }

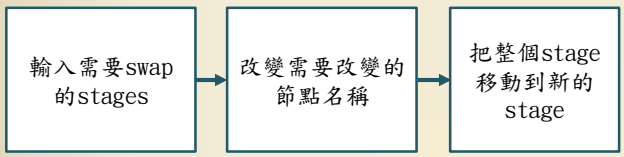
//total current
    total_i = voltage / eq_resistor[][];

//calculation voltage & current
    for(int i=0; i< stages; i++){
            for(){
                v_node[] = voltage - I_resistor * eq_resistor[][];
                I_resistor[][] = v_node[] / eq_resistor[][];
            }
    }

//print results
    for(int i=0; i< stages; i++){
        cout << left << setw() << "**";
        cout << right << setw() << "**";
    }
}
```


3. swap stage

◆思考方向:



3. swap stage

◆How to swap stage :

```
void swap_stages(int resistors[15][], int resistor_num[], string node_name[][]){
// input

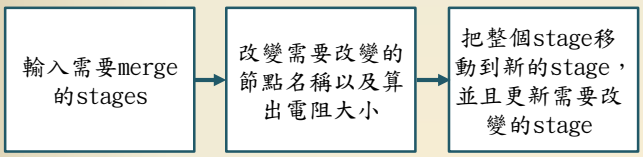
//swap
    node_nume[][] = ; //change the name of the node
    node_nume[][] = ;
    if(stage is not at the end of the circuit){
        node_nume[][] = ;
    }

    for(){ //change the stage of the node
        node_name[][] = ;
        resistors[][] = ;
        resistor_num[] = ;
    }


//cout
}
```


4. merge stage

◆思考方向:



4. merge stage

◆How to get merge stage :

```
void merge_stages(int resistors[15][], int resistor_num[], string node_name[][]){
// input

//swap
    node_nume[][] = ; //change the name of the node
    resistors[][] = ;
    if(stage is not at the end of the circuit){
        node_nume[][] = ;
    }

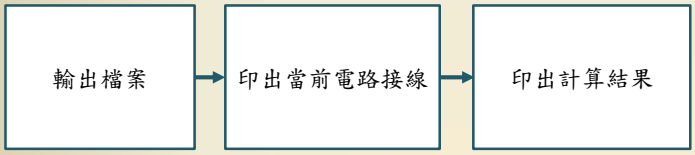
    for(){ //change the stage of the node
        node_name[][] = ;
        resistors[][] = ;
        resistor_num[] = ;
    }


//cout

}
```


5. Output file

◆思考方向:



5. Output file

◆How to get input information:

```
void output_file(int stages, int voltage, int resistors[15][], int resistor_num[],
                 string node_name[][], int V_node[], int I_resistor[][]){

//output file
    ofstream outfile();


//circuit description
    for(){
        outfile << resistors[][] << node_name[][];
    }


//calculation results
    for(){
        outfile << node_name[][] << V_node[];
    }
    for(){
        outfile << resistors[][] << I_resistor[][];
    }
}
```


5. Output file

```
***calculation results***
**************************************
**              <<voltage>>           **
**  vdd           5.000v              **
**  n1            3.000v              **
**  n2            1.000v              **
**              <<current>>           **
**  RS1_1         1.000mA             **
**  RP1_1         500.000uA           **
**  RP1_2         500.000uA           **
**  RS2_1         500.000uA           **
**  RS2_2         500.000uA           **
**  RP2_1         500.000uA           **
**                                    **
**                myspice2021         **
**************************************
```

4 spaces, 12 spaces, 16 spaces, 36 spaces, 20 spaces

Figure. 5 The format of the calculation result.



8.6 Updating Sequential Files



- Another approach is fixing the field width in advance to leave enough space for future updating.
- For example, we can set the maximum length of *name* is 11
  - `outClientFile << account << ' ' << setw(11) << name << ' ' << balance <<endl;`
  - Original File: `300 White 0.00`
  - Updated File: `300 Worthington 0.00`
  - Since the total length of this record (including space) is not changed, this update will not affect the following records.
- Although this approach does not damage other data in the same file, more redundant spaces will make the file larger.
  - Length checking is also required on the input data.

